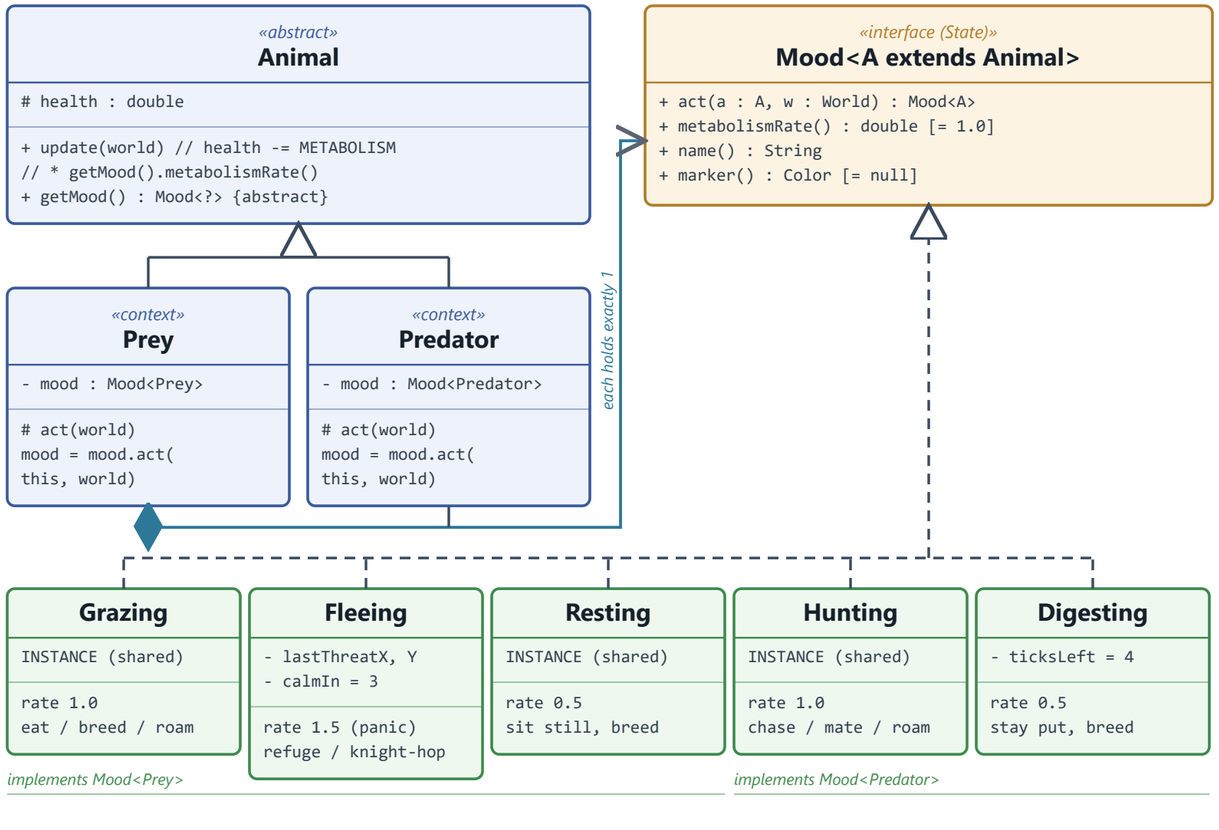
Predator and Prey Simulation — Design Patterns

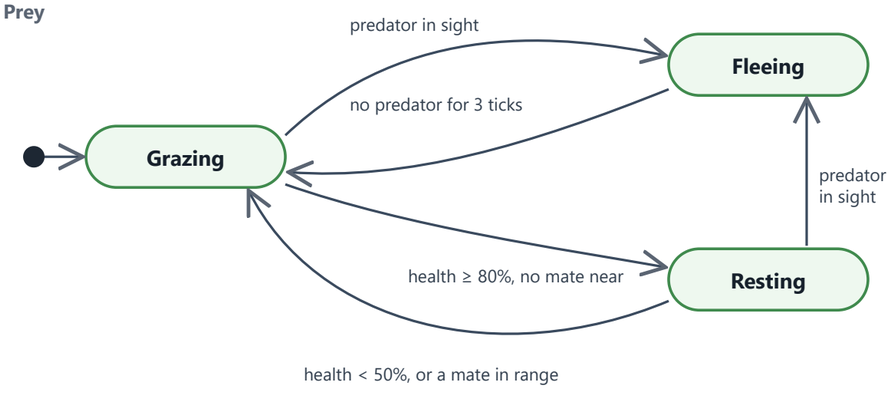
COMP2000 Final Submission · github.com/DanielZ195/COMP2000-Ecosystem

STATE — an animal's behaviour each tick is decided by its current Mood object



Transitions (each arrow is a return statement inside act(); the switch and the new mood's action happen in the same tick, so a prey never loses a step to panicking)

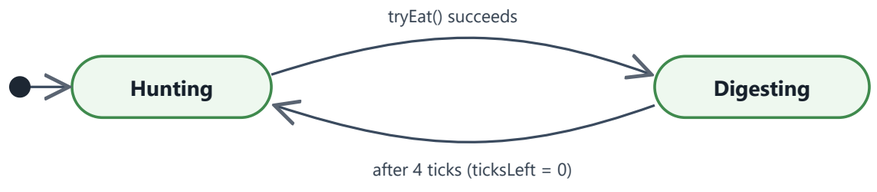
Prey



per-animal instance: remembers where the threat was last seen

shared instance; the 80% / 50% gap stops flicker every tick

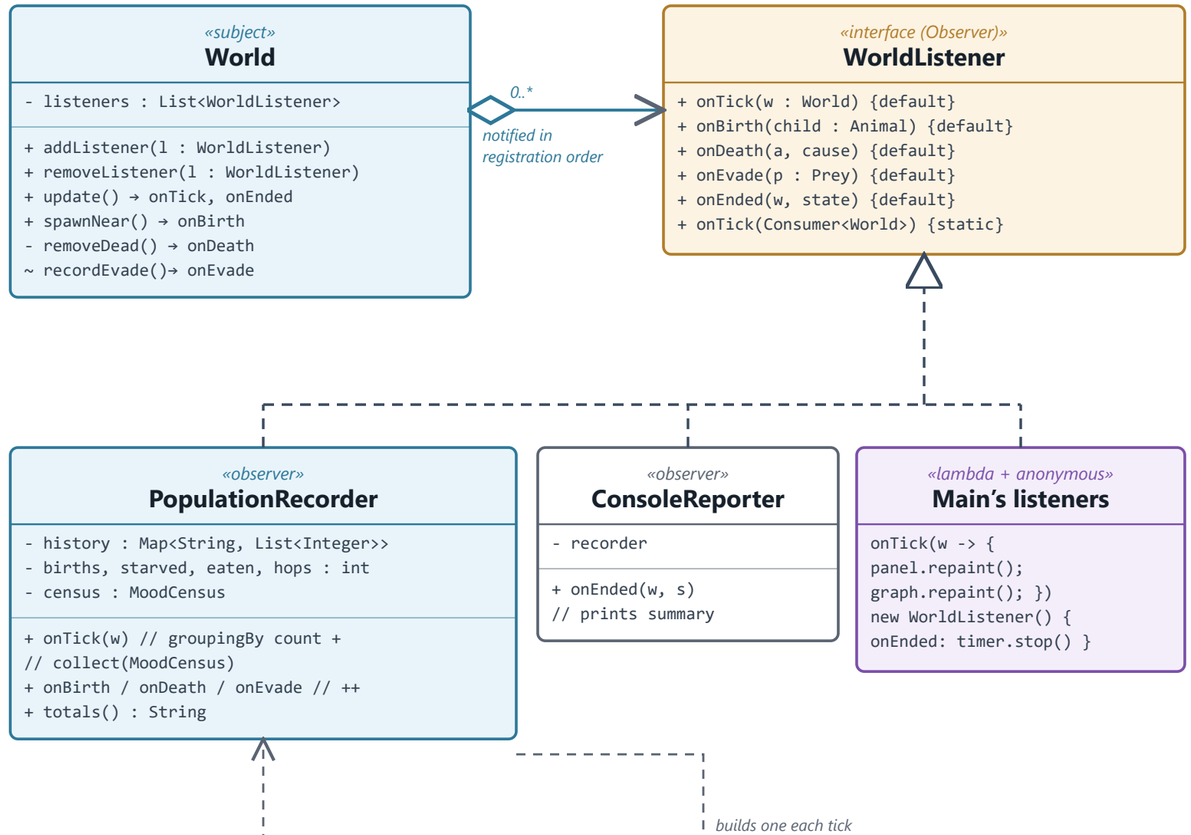
Predator



new instance per kill: the countdown belongs to one predator

Only the classes involved in the two patterns are drawn. The full class design is in design.pdf.

OBSERVER — World announces what happened; anyone interested subscribes



counts getMood().name() — where the two patterns meet

One tick, end to end



Before: Main's timer called update(), repainted both panels, checked for the end, stopped itself and printed the totals, and World kept the statistics. After: the timer only calls update(). Adding a new display or logger means one addListener() call, with no edit to World or Main's timer.

